The Journal of Risk and Insurance

© *The Journal of Risk and Insurance*, 2011, Vol. 78, No. 1, 57-82
DOI: 10.1111/j.1539-6975.2010.01402.x

Can Micro Health Insurance Reduce Poverty? Evidence From Bangladesh

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Abstract

This article examines the impact of micro health insurance on poverty reduction in rural areas of Bangladesh. The research is based on household-level primary data collected from the operating areas of the Grameen Bank during 2006. A number of outcome measures are considered; these include household income, stability of household income via food sufficiency and ownership of nonland assets, and the probability of being above or below the poverty line. The results show that micro health insurance has a positive association with all of these indicators, and this is statistically significant and quantitatively important for food sufficiency.

Introduction

Microinsurance has been described as "the protection of low-income people against specific perils in exchange for regular premium payments proportionate to the likelihood and cost of the risk involved" (Churchill, 2006, p.12). Microcredit is the provision of collateral-free small loans, especially to women, to enable them to develop household-based micro enterprises. An important objective of the microcredit movement is to break the long-standing vicious circle of poverty: from low income, to low saving, to low investment, to low income . . . and so on. The growth of microcredit is motivated by the failure of both traditional financial markets and the targeted programs of government-owned financial institutions to meet the credit needs of the

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rural poor. The importance of microinsurance emanates from the limitations of conventional loan-based microcredit programs in protecting the poor from all sorts of vulnerabilities.

Although microcredit has been shown to generate various beneficial outcomes, there is also evidence that not all sectors of the poor can benefit. One such group is those who experience severe health shocks, which reduce work capacity and investment and require a redirection of resources to the consumption of health care. Due to increased evidence that microcredit does not help the poorest of the poor, support has grown for supplementing microcredit with other services in order to improve the effectiveness of the programs. One such supplementary service is micro health insurance (MHI) to protect vulnerable people from health shocks when existing coping strategies fail. In Bangladesh, the provision of health insurance via microcredit schemes is also motivated by the lack of any health insurance facilities for the poor and the fact that the government seems unable to meet the health care needs of the rural poor.

Grameen Bank[1] (GB) has played the primary role in developing microcredit programs in Bangladesh. The organization emerged from an action research project by Professor Muhammad Yunus in 1976 examining the possibility of providing banking services for the rural poor. GB as a microfinance institution (MFI) provides a number of services including loans and savings schemes, and it added an MHI scheme in the late 1990s to protect its clients from health risks with the aim of preventing their economic downfall.

These MHI schemes may contribute to reducing poverty via improvements in health as well as reducing the health risks of the insured. Empirical evidence on whether these insurance schemes are meeting their objectives is important for policy decisions concerning expansion and replication of this type of insurance. Although there are some empirical studies on the impacts of microinsurance, especially crop insurance (Cai et al., 2010), and a large volume of literature on the role informal insurance in managing risks (Udry, 1994; Hoogeveen, 2002; Jalan and Ravallion, 1999; Kazianga and Udry, 2006; Townsend, 1994; Fafchamps and Lund, 2003; Asfaw and von Braun, 2004), to date there has been very little research on the effects of adding MHI to microcredit. Mosley (2003) examines the effects of the MHI scheme of another MFI known as BRAC (the Bangladesh Rural Advancement Committee). This study does not explicitly examine whether MHI contributes to reducing poverty, but instead it considers outcomes, such as assets, household expenditure, current saving, educational expenditure, and education level. The evidence is not conclusive as the study was conducted at a very early stage of the MHI program using a small sample. The other impact assessment studies on MHI have concentrated on process indicators, such as health care utilization and equality of access to health care in the Philippines (Dror et al., 2005), health care use and out-of-pocket expenditure in Senegal (Jutting, 2004), the utilization of health care and financial protection from health shocks in Tanzania (Msuya et al., 2007), and cost recovery in Rwanda (Schneider and Hanson, 2007). Thus, there is no good empirical evidence on whether the addition of an MHI scheme to microcredit reduces poverty.

[1]Grameen means "rural"or village" in the Bangla language.

In this article, we examine whether the addition of MHI to the microcredit programs of GB has an effect on poverty. We consider four indicators relating to household poverty status: household income, stability of household income (via food sufficiency and ownership of nonland assets), and the probability of being above or below the poverty line. We use data collected from a primary survey of 329 households in three areas where GB operates microcredit programs. The areas are distinguished according to their experience of MHI: areas with at least 5 years experience of MHI, those with 2 years or less experience, and those where MHI is not available. Our evidence is based on econometric analysis of the impact of placement of MHI. The study finds a positive association between MHI and all of the poverty indicators, and this is statistically significant in the case of food sufficiency. In our discussion, we consider some of the possible explanations for this finding.

This article is organized as follows. The second section provides some background on microcredit, health, and MHI programs in Bangladesh. The third section illustrates the conceptual framework explaining the possible causal pathways through which adding MHI to microcredit can affect the poverty status of households. The fourth section describes the econometric methodology. The fifth section reports the results and the sixth section discusses them. The seventh section offers some conclusions.

BACKGROUND

There has been a massive expansion in Grameen-type microcredit in Bangladesh since the 1990s. The Palli Karma Sahayak Foundation is an apex organization for microcredit funding and capacity building; according to their figures, in December 2005 there were around 700 MFIs in Bangladesh with over 33 million members. The main players in the microfinance sector are GB, BRAC, ASA, and Proshika. Some government departments and commercial banks also provide microcredit.

Although microcredit generates various beneficial outcomes (Pitt and Khandker, 1998; Pitt et al., 2003; Khandker, 2005; Gertler et al., 2009), there is evidence that households that experience severe health shocks are unable to reap much benefit from microcredit. Illness, especially chronic illness, of household members is the main reason that GB borrowers fail to rise above the poverty threshold (Todd, 1996; Wright, 2000). This is because ill health reduces work capacity and investment due to a redirection of resources to consumption of health care. Thus, welfarists stress the value of adding auxiliary services to microcredit in order to improve the effectiveness of the programs (Dichter, 1996; Jani and Pedroni, 1997; Woller, Dunford, and Woodworth, 1999; Bhatt and Tang, 2001; Woller and Woodworth, 2001).

Insurance may protect vulnerable people from risks and shocks. However, traditional health insurance markets are almost entirely absent in the rural areas of Bangladesh. There is no social health insurance scheme even in the formal sector in Bangladesh, and the government has not been able to meet the health care needs of the rural poor despite having a well-established health care delivery network.

The constitutional commitment of the government of Bangladesh is to provide basic medical care to all of its citizens; it has invested substantially since independence to develop the health infrastructure and to strengthen health and family planning services with special attention to the rural population. Providing primary health care

to attain "Health for All" is the major thrust of the health program. There is a three-tier mechanism for providing health care in rural areas: (1) domiciliary services by a Health Assistant and Family Welfare Assistant at the household level, (2) Health and Family Welfare Centers at the union level,[2] and (3) Upazila Health Complexes (UHCs) at the subdistrict level. UHCs provide both outpatient and inpatient services, including maternal and child health and family planning; they are the main center for implementing the Essential Services Package, which was designed to attain Health for All. In addition to public provision of health care, there is a large private sector in Bangladesh, that includes both not-for-profit and for-profit organizations; the former is relatively small and run by NGOs, MFIs, and charitable institutions.

Despite this infrastructure for health care delivery, the government has largely failed to meet the health care needs of the rural population, and this is due mainly to supply-side constraints: first, problems in retaining doctors in UHCs due to poor working conditions; second, a lack of proper input mix and skill mix due to underresourcing and recruitment problems; third, unfriendly and unapproachable behavior of the health care providers, which discourages contact from the local population; and finally, the charging of unofficial fees. Thus, although there is underutilization in many UHCs, the majority of patients seek health care from private providers, especially from informal providers who often have no formal medical qualifications (BBS, 2006).

In order to expand into areas of social protection not covered in conventional loan-based microfinance and to help supplement poor government provision of health care, GB set up an MHI scheme to protect its microcredit clients from health risks in order to ameliorate the economic risks they face. The key features of the scheme are shown in Table 1. GB sells annually renewable prepaid insurance cards to its clients and offers primary health care directly from its own health centers. The service comprises mainly curative care and maternity and child health care. Noncard-holders can seek health care from these health centers but they pay a higher charge for the services compared to cardholders. Following GB, some other MFIs in Bangladesh have also introduced MHI schemes with similar objectives. In terms of the type of insurance provided, the GB scheme is largely prepayment for health services rather than a program that covers most catastrophic health costs. Thus, it may be the case that the government provision, while inadequate, does provide some coverage for the more catastrophic needs while failing to provide basic health services.

CONCEPTUAL FRAMEWORK

An insurance contract is mainly aimed at reducing the risk faced by the person who buys it. The theory of demand for insurance is based on expected utility (EU) theory (von Neumann and Morgenstern, 1944).

An individual purchases health insurance if her EU with insurance is greater than that without insurance.[3] Risk-averse individuals (those with concavity of the utility

[2]There are 482 "upazilas" (or subdistricts) in Bangladesh, which are the lowest level of administrative government. Each upazila is subdivided into smaller local "unions."

[3]Assume that there is a probability, ρ, that an individual will become ill and spend L on medical care. Otherwise, the consumer will purchase full insurance coverage, I, for the actuarially fair premium of $R = \rho L$. Thus, $EU_0 = (1 - \rho)U(Y) + \rho U(Y - L)$ is EU without insurance and

TABLE 1
A Picture of the Key Features of the MHI Scheme of GB

Key Features	Grameen Kalyan of GB
Type of organization	• Insurer as well as a health service provider.
Area coverage	• Dec. 2008: 39 unions of 25 subdistricts of 10 districts. Program identical in all areas.
Mechanisms for providing health services	• Mainly through static clinics.[a] • Also via mobile clinics, satellite clinics, and domiciliary visits (by the health workers). • Limited emergency services through static clinics. • Referral services via agreements with some hospitals.
Technical staff	• MBBS (Bachelor of Medicine and Bachelor of Surgery) doctor. • Female DMF (Diploma in Medical Faculty). • Office manager, female paramedic, laboratory technician, six community health assistants. • Trained Traditional Birth Attendants (TTBA) trained by Grameen Kalyan.
Categories of services covered	• Preventive care including antenatal care (ANC). • Curative care (mainly outpatient care). • Basic diagnostic services including ultrasonography. • Some health promotion activities.
Type of curative services provided	• Essential Services Package (ESP) including safe delivery.
Enrollment status	• Voluntary.
Proof of enrollment	• Insurance card.
Affiliation unit	• Household. Program identical for all enrollees.
Eligibility criterion	• GB member households or any villagers living within 8 km of health center.
Access of noninsured households to curative care	• Yes.
Premium	• Annual premium (covering up to six members)[b] ■ GB member family: TK.120 (US$1.74) ■ Non-member family: TK.150 (US$2.17) • Average daily wage ■ Male: TK.90 (US$1.30) ■ Female: TK.60 (US$0.87)
Benefit package for the cardholders	• Copayment: ■ Medical consultation, cardholder: TK.10 (US$0.14) ■ Medical consultation, non-cardholder: TK.25 (US$0.36) • Discount: basic medicine[c] on MRP: 25%; pathological tests on listed price: 30–35%; referred consultation visit: 50%. • Hospitalization benefit: Annually up to 2000TK (US$29) for a family. • Free: Annual basic check-up for head of the family, immunization against six diseases, domiciliary visits by health assistants.

(*Continued*)

TABLE 1
Continued

Key Features	Grameen Kalyan of GB
Additional health services package	• School health package, Eye Mega Camp for cataract operations, and regular cataract operation programs.
Financing mechanism	• Initial endowment fund of Grameen Bank. • Revenue generation from premiums and copayment.
Cost recovery rate	• 100% (including managerial and overhead costs of Regional and Head Office) in most of the old health centers.
Joining mechanism	• At weekly microcredit meetings for GB members. • Via health visitor domiciliary visits for GB members. • At health centers; for GB members or nonmembers.

Source: Hamid, Bhuiyan, and Sultana (2005), Ahmed et al. (2005), and various official documents of SSS and Grameen Kalyan of GB.
[a]Static clinics are the normal clinics where patients come to seek health services, in contrast to "satellite clinics" where the physicians/nurses visit a location regularly to provide care.
[b]TK.20 is charged for each additional member.
[c]Basic medicine: 15 essential medicines are listed in the schedule of Government of Bangladesh.

function of wealth) demand health insurance to protect themselves from the financial losses of illness. The underlying assumption is that individuals prefer certain losses over uncertain ones of the same expected magnitude (Friedman and Savage, 1948; Arrow, 1963).[4] Thus, "the purpose of any insurance policy is to convert an uncertain, but potentially large, loss into a certain, small loss" (Newhouse, 1978, p. 19).

Several factors affect the demand for health insurance of an individual: degree of risk averseness, probability of becoming sick, magnitude of the loss, the price of insurance, and income. In traditional health insurance, payment is made upon either state of health (disease contingent contract) or reimbursement of health expenditures (Besley, 1989); the latter is more common in practice. Moral hazard (both consumer and provider) and adverse selection arising due to asymmetry of information are the two common causes of nonprofitability of health insurance.[5] Some risk-sharing tools (such as deductibles, copayments, and coinsurance) are used to reduce consumer moral hazard.

$EU_i = (1-\rho)U(Y-R) + \rho U(Y-L+I-R)$ is with insurance, where Y is the disposable income. A utility-maximizing risk averse individual will purchase insurance if $EU_i > EU_0$.
[4]Some evidence against the conventional theory exists. Empirical studies have shown that there is no link between the shape of the utility function of wealth and the attitude toward risk (Kahneman and Tversky, 1979; Tversky and Kahneman, 1981, 1986). Nyman (2003) claims that "consumers prefer an uncertain loss to a certain loss of the same expected magnitude" (Nyman, 2003, p. 1).
[5]Consumer moral hazard (both *ex ante* and *ex post*) raises the demand for curative care. Adverse selection arises due to offering a premium based on averaging the risks of the low- and high-risk groups instead of offering separate premium schedules for them. Thus, the low-risk group may be discouraged from purchasing insurance.

Unlike the traditional three-party insurance health insurance system (the insurer, the insured, and health service providers), the MHI schemes of Bangladesh are a two-party system—the insurer directly provides health services. The service package includes both curative (mainly primary) and preventive health care including health education services (see Table 1). Hence, demand for MHI may stem from demand for health care directly as well as from a preference to smooth consumption. In addition, these schemes are usually tied up with microcredit programs. Thus, in comparison to traditional health insurance, the demand for these schemes may be more influenced by noneconomic factors. Provision of preventive care, including health education, checks the *ex ante* moral hazard while coinsurance checks *ex post* moral hazard. Adverse selection is not a major problem for these schemes because the packages are focused on the poor, who are usually a high-risk group.

The MHI schemes have limited scope to cover large losses, and due to very high coinsurance the clients may not receive a true sense of security from these schemes (which may result in a high dropout rate); however, whatever its risk-avoidance capacity, MHI is a way of accessing quality health care, which is imperative in any developing country like Bangladesh.

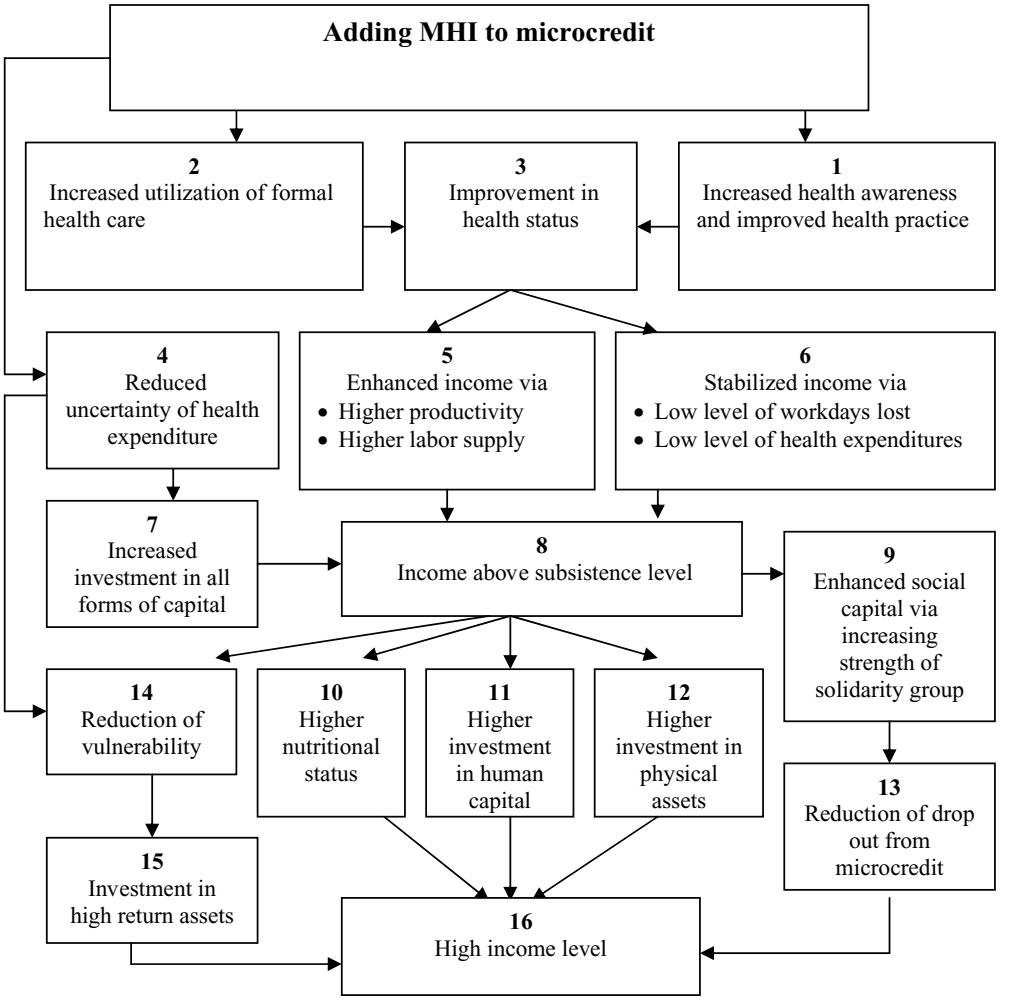
A MHI scheme offering both curative and preventive health care and health promotion services may increase the health status of the participating households via increased health awareness, improved health practices, and increased utilization of formal health care (see Figure 1). There is little empirical evidence for these effects in countries like Bangladesh but evidence based on health maintenance organizations (HMOs) in the United States suggests that there are causal effects from HMOs to a higher incidence of preventative care (Rizzo, 2005), and increased use of health care services (Shin and Moon, 2007). Improved health status may lead to higher productivity, higher labor supply, fewer workdays lost, and reduced health care expenditure. The first two can be regarded as "augmenting" effects and the latter two as "stabilization" effects.

Under augmenting effects it is assumed that improved health status leads to an increase in the efficiency of labor supply as well as actual labor supply, which increases productivity, and thereby earnings, in the microenterprise.[6] The stabilization effects can be elaborated as follows. First, improved health status leads to lower income loss through reducing the workdays lost due to illness. Second, it reduces both direct and indirect health expenditures because health care utilization may be reduced due to improved health status.[7] Thus, the household may maintain health care expenditure from regular income or savings. This may help the household avoid borrowing or

[6]In addition, as in most cases the microentrepreneurs are women, they also tend to take care of sick children and adults in the household; therefore, increased health status of household members reduces the time spent caring and thereby increases the supply of effective labor.

[7]It is worth mentioning that this may happen only if there is a substantial improvement in health status, which may not occur in the short run. Thus, health care expenditure may not fall in the short run; rather, it may increase because insurance provides incentives to use more health care.

FIGURE 1
Potential Effects of Adding MHI to Microcredit



selling of productive assets to meet the medical needs of the household,[8] which may prevent falling incomes earned from microenterprise.

In addition, the supplementation of microcredit with MHI may reduce the uncertainty of health care expenditure and thus raise investment in all forms of capital. If the household is insured against health risk, it may invest in high-return riskier assets because it does not need to retain cash or to hold easily liquidable assets for precautionary purposes. It is worth mentioning that Kochar (2004) finds, from a study

[8]Poor households sell various household assets like livestock, land, and grain and take out loans to cover the financial costs of illness (Sauerborn, Adams, and Hien, 1996).

in rural Pakistan, that overall savings of households rise in the expectation of future illness of adult males, but investments in productive assets decline.

Thus, these two channels (improvement of health status and reduction in uncertainty regarding health care expenditure) may result in higher household income. When income rises above subsistence level, the household may increase investment in nutritional improvement, human capital, and physical capital. This may also reduce vulnerability, which may in turn lead to investment in high-return assets. Moreover, surplus income may enhance social capital through increasing the strength of the solidarity of the group members, which may reduce the probability of dropout from the microcredit program. These effects together may lead the household to attain a higher income state in the long run, and thus it reduces the probability of a household being below the poverty line.

Another issue that we should consider here is the potential for informal risk sharing or insurance, which is an important way in which people in less developed regions deal with risk (see surveys by Alderman and Paxson, 1994; Morduch, 1995; Townsend, 1995; Fafchamps, 1999; Dercon, 2002). Informal insurance can take several forms, including drawing on savings, running down crop stores, selling physical assets, transfers within family networks, or reciprocal borrowing arrangements within communities. Some studies have warned that formal insurance schemes might crowd out these informal mechanisms, but in his review, Mordoch (1999) concludes that informal insurance mechanisms are typically weak and that crowding out is unlikely to be a significant problem. Gertler and Gruber (2002) evaluate households' ability to cope with the costs of illness in Indonesia where reliance is on informal risk sharing. They find that households can insure against around 70 percent of common health shocks but can protect their consumption levels against only around 30 percent of the illnesses that have a significant effect on long-term performance. Fafchamps and Lund (2003) find that in the rural Philippines informal insurance has limited ability to cope with even mild illnesses and that a household's social network is very important in that households with friends who are well-off fare better. Cai et al. (2010) also point out that studies that compare outcomes for households with and without formal insurance may be liable to underestimate the effects of formal insurance because of the existence of informal risk sharing between households that do have formal insurance and those who do not. Similar arguments are made by Angelucci and De Giorgi (2009). We discuss the implications of these arguments for our empirical analysis in the Methodology section .

METHODOLOGY

Data

We collected primary data from a quantitative household survey through a cross-section design in 2006. We chose to study the MHI component of GB because this MFI operates on a large scale in Bangladesh. In June 2006, MHI was being operated in 32 of GB's branches, of which 14 branches had the scheme for at least 5 years and the remaining 18 had MHI for 2 years or less. The GB microcredit and MHI programs are identical across the areas that we study. A description of the key features of the GB MHI scheme is given in Table 1. In order to construct a meaningful study design that would enable us to test causal relationships, GB branches were stratified into

three distinct types: (1) GB—branches with at least 5 years experience of MHI; (2) GB2—branches with 1 or 2 years experience; (3) GB3—branches without MHI.

A multistage sampling technique was applied. One GB branch was randomly selected from each of GB1 and GB2; they are Madhabpur and Pakutia, respectively. Madhabpur is located at Singair upazila (subdistrict) of Manikgonj district and Pakutai is at Nagarpor upazila of Tangail district. One GB branch (Joy Mantap) was selected from GB3. This branch was purposively selected from the same upazila (Singair) as Madhabpur in order to make a meaningful comparison group. Note that there are eight branches of GB in Singair upazila and an MHI scheme has been operating in its three unions (Madhabpur, Shaharil, and Jamsaha) since 1996. Among the remaining five unions where GB has not yet placed its MHI scheme, Joy Mantap is adjacent to Madhabpur and they are connected by road. It is worth mentioning that GB has decided to introduce its MHI scheme into Joy Mantap in the near future, which may reduce program placement bias in this design. It also may be noted that microcredit was placed at Madhabpur and Joy Mantap in 1983 and at Pakutia in 1986. An MHI scheme was placed at Madhabpur and Pakutia in 1996 and 2004, respectively.

A list of all the villages holding at least one female center of GB was prepared for each selected area.[9] In the second stage, two villages from each area were selected randomly. Thus, a total of six (3×2) villages were selected. In the third stage, two female microcredit centers were selected randomly from each village where more than two microcredit centers existed. Some villages had only one credit center, in which case it was selected. Note that each credit center consists of 40–50 microcredit members and they usually live in a particular area of the village. A list of current GB microcredit member households was made in each selected GB loan center, using information obtained from respective branch offices of GB. We attempted to interview all the eligible member households in the selected microcredit centers of each sampled village in order to control for sample selection bias.

We used a set of questionnaires to collect both household-level and village-level information. We finalized the survey instruments after incorporating comments and suggestions from a number of experts who were sent a preliminary draft of the questionnaires, as well as feedback from two rounds of piloting. We employed a team of six female surveyors comprising four field investigators, one field supervisor, and one quality controller to conduct the survey. All members of the survey team had completed their masters degree and were experienced enough to conduct the field survey. We provided them with 5 days training on the purpose and objectives of the study and on the use of the survey instruments.

Female microcredit members were the key respondents; however, some questions (income, expenditure, loans, assets, etc.) were asked of household heads and these are predominantly male. The village-level information was collected from the offices of the Union Councils and key informants. The data were entered in SPSS v10 and converted to Stata v9 for analysis.

[9]Since about 96 percent of existing GB members are female, we selected only the female microcredit centers.

Method

One of the major challenges in estimating the impact of a program (like MHI) using nonexperimental data is to deal with endogeneity caused by heterogeneity in unobservable individual characteristics of the participants and nonparticipants, which may influence both the decision to participate in the scheme and the outcome. Random assignment of individuals into a treatment group and a control group can balance the heterogeneity in unobservable individual characteristics and thus eliminate bias. However, random assignment is not possible here; hence, it is necessary to choose an empirical model, which can control for endogeneity.

Following the empirical literature on health insurance and microcredit (Pitt and Khandker, 1998; Nanda, 1999; Waters, 1999; Yip and Berman, 2001; Jowett, Contoyannis, and Vinh, 2003; Trujillo, 2003; Khandker and Faruqee, 2003; Jutting, 2004; Jowett, Deolalikar, and Martinsson, 2004), we consider a *structural* equation and a *reduced-form* equation to estimate the outcomes of the interest.

In the structural equation:

$$y_{ij} = X_{ij}\beta + A_{ij}\delta + \varepsilon_{ij}, \tag{1}$$

y_{ij} is the outcome of interest (household income, stability of household income, or probability of being in poverty for household i in village j).[10] X_{ij} is a vector of observed individual, household and village characteristics (education, age, duration of membership in MHI, and so on). A_{ij} is a binary variable where $A_{ij} = 1$ if household i of village j participates in MHI and $A_{ij} = 0$, otherwise; and ε_{ij} is the stochastic error term. The estimate of δ will give the unbiased effect of MHI on the outcome y only if A_{ij} is an exogenous variable.

In the reduced form (or participation) equation

$$A_{ij} = X_{ij}\beta + Z_{ij}\phi + \mu_{ij}, \tag{2}$$

X_{ij} is as defined in Equation (1), Z_{ij} is a distinct set of household or village characteristics that affect only participation in the scheme (A_{ij}) but not the outcome (y_{ij}) conditional on A_{ij}, and μ_{ij} is the stochastic error term. Endogeneity arises when A and ε (or ε and μ) are correlated, resulting in biased estimates of δ.

While conditioning on participation is the commonly used method for analyzing the causal effect of health insurance as evidenced in the literature, this method is not appropriate for our research for a number of reasons. First, MHI is likely to produce substantial spillover effects because the scheme offered by GB is different from a traditional three-party (the insurer, the insured, and health service providers) health insurance system. As well as prepaid insurance cards, GB MHI provides health care directly to their clients via health centers. In addition, it offers health promotion services to all the microcredit members living under the catchment areas of the health

[10] The exact specification of the model (linear, binary probit, or ordered probit) depends on the nature of the outcome.

center regardless of their participation in MHI.[11] Moreover, the uninsured can also seek health care from the GB health centers by paying the standard fees. The issue of informal risk sharing is also incorporated into the idea of spillover effects. Some uninsured households may benefit from MHI via informal risk sharing with insured households, for example, through borrowing or other transfers. If these spillover effects are not taken into account, the impact of MHI will be severely underestimated. Second, there was a very high enrollment rate in MHI at GB1 (96 percent). Since there are very few observations on the nonparticipant group, the estimation of Equation (1) may not give sensible findings.

Thus, in what follows we focus on outcomes conditioned on *placement* of the program rather than *participation* in the program. The available empirical literature on this method falls mainly into two groups. One group has compared the program group with a comparison group where the program was in the pipeline (Chase, 2002; Galasso and Ravallion, 2004). The other group has compared the program group with a comparison group where the program has not been placed at the time of survey (Amin, Li, and Ahmed, 1996; Hadi, 2002). The placement model can be written as follows

$$y_{ij} = X_{ij}\beta + P_{ij}\lambda + \varepsilon_{ij}, \tag{3}$$

where y and X are as defined previously, and P represents program status where $P = 1$ if the household is drawn from a program area where MHI has been operating for at least five years (GB1), and $P = 0$ if the household is drawn from the comparison area where MHI was not placed at the time of survey (GB3).[12] The estimate of λ measures the average treatment effect of MHI. Measuring *average treatment effect* rather than *average effect of treatment on the treated* is sensible because nonparticipants may also benefit from the availability of MHI due to spillover effects. For example, to the extent that informal insurance mechanisms arise in the program areas as an externality to the MHI scheme, this will be included in our estimate of the average treatment effect.

The major concern in this method is to control for program placement bias (the particular features that attract an organization to place the program in an area) and geographical heterogeneity, which may affect the outcomes. Social programs like MHI are rarely placed randomly; rather, placement depends on both demand and supply-side factors. The best way to ameliorate the effects of program placement bias is to select a suitable comparison area, and to do this we took the following measures: first, selecting the program area (GB1) and control area (GB3) from the same small geographical region to reduce geographical heterogeneity; second, choosing the control area where there is potential for placing the program in the near future to reduce the supply-side bias of program placement; and third, conducting a survey to see whether eligible households in the control area would be willing to be insured if MHI was placed in their village, to control for demand side bias.

[11] It is also reasonable to expect that the benefits of health promotion will extend beyond microcredit members to any of local people.

[12] Note that GB2 does not feature in the evaluation, which compares treatment group (GB1) with control group (GB3). GB2 is included only to improve the richness of the description of MHI in Bangladesh.

In addition, we used a similar method to select the households from the program area and control area. It should also be noted that we conducted the survey on female microcredit members, and this group is likely to be relatively homogeneous across the program and control areas. Informal insurance mechanisms may exist in both our program and comparison areas. As all our sampled households are (group-based) microcredit members, there is likely to be some networking among the group members in all areas. However, the presence of microcredit for more than a decade in all areas may reduce the need for informal insurance. Apart from any difference arising from the MHI scheme itself (therefore only available in the program areas and incorporated in the average treatment effect), we have no reason to expect the extent of informal risk sharing to be substantially different between areas. We explore this in our empirical work described in the fifth section.

Poverty Measurement Indicators

In order to gain a deeper understanding of the impact of MHI on poverty status, we consider four outcome measures that relate to poverty: household income, household nonland assets, food sufficiency, and the probability of being above or below the poverty line. These outcomes can be related to the mechanisms illustrated in Figure 1. Household income relates directly to boxes 5, 8, and 16, which reflect enhanced income via health improvements. Food sufficiency and ownership of nonland assets relate to boxes 6 and 7, where income is stabilized via reduced workdays lost and lower health care expenditure leading to increased investments in all forms of capital;[13] in addition, food sufficiency contributes directly to higher nutritional status (box 10). The final poverty measure is a direct reflection of boxes 8 and 16 (higher income, which crucially may then be above subsistence level).

Clearly, we are using outcome measures at one point in time, derived from a cross-section survey, to reflect the results of the dynamic processes outline in Figure 1. We might expect that some of our indicators will react more quickly to the potential beneficial effects of MHI than others; for example, it seems reasonable to expect that food sufficiency and per capita income may be the first to change whereas accumulation of nonland assets and increasing incomes above the poverty line may take longer to achieve.

Total household income is constructed from a series of questions on all sources of income of all members of the household in the last 12 months.[14] In the analysis we use the log of age- and sex-adjusted per capita income.[15] Household nonland assets and food sufficiency are used to proxy for the stability of income. Food sufficiency is an

[13]Food sufficiency is also highly affected by local factors such as weather, river erosion, and political stability. We collected data on the weather from all the sampled villages. We did not find any difference between GB1 and GB3. In addition in Bangladesh, political stability tends to be a national phenomenon.

[14]We have measured gross income rather than net income (income retained after deduction of production costs) because the respondents find it difficult to state the costs of production accurately.

[15]The weights used to construct age- and sex-adjusted household size are: 1.0 for males aged over 18 years; 0.9 for females aged over 18; 0.94 for males aged 13–18, 0.83 for females aged over 13–18; 0.67 and 0.52 for a children aged 7–11 and 4–6, respectively; 0.32 for toddlers

appropriate proxy because it varies with fluctuations in income level. Also, ownership of nonland assets is suitable because poor people usually depend on selling these liquidable assets to cope with any income shock. Respondents were asked to classify their household food consumption for the year as either: in deficit for the whole year, sometimes in deficit, neither in deficit nor surplus, or in surplus for the whole year. There were very few observations in the "deficit for the whole year" category, so this group is merged with the "sometimes in deficit" group to create a 3-point ordered food sufficiency scale, where 0 = in deficit, 1 = neither deficit nor surplus, and 2 = in surplus. Information on ownership of nonland assets was constructed from a series of questions about all nonland assets and their estimated current market price. We use the current value of all household nonland assets whether they contribute to household income or not. As for per capita income, we have used age- and sex-adjusted household size to measure per capita nonland assets.

We use the "cost of basic needs" approach to measure the poverty line; here, the cost of a normative bundle of goods needed to meet minimum nutritional requirements plus the costs of nonfood basic need items is defined as poverty line expenditure. Following Ravallion and Sen (1994), we use 2,112 calories and 58 of grams protein as the minimum daily calorie requirement of an adult for the food poverty line and then take 35 percent of the food poverty line income as the appropriate nonfood allowance. We then define households as falling below (= 0) or above (= 1) the poverty line according to these criteria.

FINDINGS

Descriptive Statistics

A total of 329 households were surveyed, of which 136 were from GB1, 85 from GB2, and 108 from GB3. The overall response rate in the survey was 73 percent (75, 68, and 73 percent in GB1, GB2, and GB3, respectively).[16] There was little variation in response rates between the insured and the uninsured. The participation in MHI among the households interviewed in the survey was 82 percent (96 percent at GB1 and 59 percent at GB2). One of the reasons for the low participation rate in GB2 may be the shorter period of operation. It is noteworthy that 31 out of 85 respondents in GB2 could not state whether they were insured or not; this is likely to be because they were in the process of being enrolled. Respondents in GB3 were asked whether they would like to enroll in MHI if it was placed in their area. The willingness to participate in MHI at GB3 was 98 percent, which is similar to the actual enrollment rate in GB1 (96 percent).

There is no significant difference in background sociodemographic features, apart from duration of membership in microcredit, between the program area (GB1) and the control area (GB3) (see Table 2). Also, there is no difference in main sources of income, general infrastructure, and occurrence of natural calamities. In addition, household health care expenditure and workdays lost do not differ significantly between GB1

aged 1–3; and 0.05 for infants. These weights are based on a South Indian dietary survey and are also used by Townsend (1994).

[16]The main reason for nonresponse is that we could not interview some households due either to absence of the key respondent or unwillingness to take part in the survey; this was largely because the survey was conducted during harvesting season where a number of household members are away from home for long periods. In order to reduce this problem one repeat

TABLE 2
Sociodemographic Characteristics of the Sampled Population

Key Features	GB1	GB2	GB3	Difference Between GB1 and GB2	Difference Between GB1 and GB3
Membership rate (%) of households in GB microcredit program	34	32	33	0.02	0.01
Average duration (in year) of membership in microcredit program	9.62 (6.95)	6.42 (6.05)	6.84 (6.25)	3.19***	2.77***
Average length of membership in MHI program	6.01 (2.96)	1.00 (0.20)	-	5.02***	–
Average education level (male)	2.97 (3.16)	4.21 (3.86)	2.97 (3.43)	−1.24***	0.00
Average education level (female)	2.28 (2.73)	2.80 (3.33)	2.23 (2.95)	−0.51*	0.05
Average household size	4.84 (1.39)	4.90 (1.72)	4.62 (1.3)	−0.06	0.23
Male population (%)	50.91	44.84	51.50	0.06**	−.01
Female population (%)	49.09	55.16	48.50	−0.06**	0.01
Average age	25.37 (18.94)	27.45 (20.19)	26.74 (19.29)	−2.08*	−1.38
Average wage (male) in US$	1.3 (0.90)	1.4 (0.93)	1.2 (0.97)	−0.01	0.01
Average wage (female) in US$	0.85 (0.81)	0.86 (0.76)	0.83 (0.80)	−0.01	0.02
No. of MFIs working	5	6	5		
Workdays lost per sick person	4.10 (6.41)	5.34 (8.79)	4.24 (6.43)	−1.24	−.14
Health care expenditure (US$) per sick person	31.68 (66.24)	33.60 (54.36)	25.72 (37.28)	−1.91	5.96

Notes: The *t*-statistics are used to test the significance of the difference between the study group and the control cohort. ***indicates significant at the 1% level, ** significant at the 5% level, *significant at the 10% level. Magnitudes in parentheses are standard deviations.

and GB3, which may cast doubt on some of the causal pathways described in Figure 1. There is a significant difference in most of these features (duration of membership in microcredit program, male and female education, and age and sex structure) between GB1 and GB2. As per the design of the study there is a significant difference in duration of membership in MHI between these two areas. At GB1, about 64 percent of the insured had at least 5 years experience of MHI (the average experience is about 6 years) and at GB2, 96 percent had only 1 year experience.

Almost all the microentrepreneurs were aged between 17 and 64 years irrespective of the sample areas. Most (88 percent) of them were of reproductive age (15–49 years).

visit was made to absent households. Respondents were not pressured or offered motivation to take part in the interview.

More than 90 percent were married in all the sampled areas. More than 60 percent of the microentrepreneurs, irrespective of the sample areas, had no formal education. But the formal education rate was higher in GB2 (38 percent) compared to other areas (25 percent). The majority (more than 65 percent) of the microentrepreneurs in GB2 and GB3 had household-based self-employment,[17] and this was around 44 percent for the microentrepreneurs in GB1. A good number (28 percent) of microentrepreneurs of this area had small businesses. Some microentrepreneurs (18 percent, 22 percent, and 17 percent for GB1, GB2, and GB3, respectively) were not directly involved in any economic activity. Rather, they gave the money borrowed from MFIs to someone else either within or outside the household. Microentrepreneurs were themselves the household head in some cases (15 percent, 14 percent, and 10 percent at the GB1, GB2, GB3, respectively), but a vast majority (more than 80 percent) of households were male headed. Like microentrepreneurs, the majority (more than 65 percent) of the household heads had no formal education in GB1 and GB3; this was around 42 percent for the household heads in GB2. Small business, farming, and day labor were the major occupations of household heads in all the areas.

In relation to informal insurance, there is evidence that loans and transfers within networks play a strong role in risk sharing (Fafchamps and Lund, 2003) or at least some partial role (Hoogeveen, 2002; Jalan and Ravallion, 1999). We included a question on the source of financing of health care giving five options: (1) out of regular current income, (2) drawing from accumulated saving, (3) sale of assets, (4) borrowing/loan, and (5) others. The last two options may be used as a proxy of informal insurance. There were no significant differences between the sources across areas. Slightly less expenditure came from current income in GB1 (92 percent) compared to GB2 and GB3 (95 percent in both cases), which is consistent with the long-term establishment of MHI in GB1. The overwhelming majority (96 percent) of households in all areas finance health care mainly from the first two sources, which suggests little reliance on informal risk sharing. This finding is consistent with that of Ahsan, Hamid, and Barua (2010) based on a survey of 4,000 households in rural Bangladesh.

Turning to our poverty outcome measures in Table 3, there is substantially higher mean per capita income and per capita nonland assets in GB1 compared to GB3 or GB2. According to the "cost of basic needs" method, the per capita annual poverty line expenditure was US$155.77 in GB1 and US$156.05 in GB3. A substantially higher proportion of households are above the poverty line in GB1 compared to GB2 and GB3, and in both cases this difference is significant at the 10 percent level. The data also show that the majority (more than 60 percent) of households do not have a food deficit irrespective of MHI placement status. Although food surplus (and absence of deficit) is higher in GB1 compared to GB2 or GB3, the differences are not significant at any conventional level.

Multivariate Analysis

The regression results are presented in Table 4. The estimation technique is chosen depending on the form of the dependent variable. Models (i) and (ii) are estimated by OLS for the continuous variables log per capita income and log per capita nonland

[17]Note that livestock fattening, poultry feeding, farming, tailoring, etc. fall in this category.

TABLE 3
Poverty Indicators by GB Area

	GB1	GB2	GB3	Difference Between GB1 and GB2	Difference Between Difference
Mean per capita income US$	342.7 (174.6)	260.1 (135.0)	240.3 (121.4)	82.1***	102.3***
Mean value of per capita nonland assets US$	223.5 (203.4)	142.0 (106.2)	158.2 (127.1)	81.5***	65.3***
Percentage of households living above poverty line	80.88 [110]	70.37 [76]	69.44 [75]	10.51*	11.44*
Food sufficiency level					
In deficit whole year	2.2 [3]	7.1 [6]	5.6 [6]	−4.9	−3.4
Sometimes in deficit	22.1 [30]	31.8 [27]	32.4 [35]	−9.7	−10.3
Neither in deficit or surplus	49.3 [67]	48.2 [41]	53.7 [58]	1.1	−4.4
Surplus	26.5 [36]	12.9 [11]	8.3 [9]	13.6	18.2

Notes: The *t*-statistics are used to test the significance of the difference between groups. ***indicates significant at the 1% level and * significant at the 10% level. Figures in brackets are the number of observations. Figures in parentheses are standard deviations.

assets, since the histograms of the log-linear data for both of these variables are approximately normal.[18] An ordered probit model is estimated for food sufficiency level (iii), where food deficit = 0, neither deficit nor surplus = 1, and surplus = 2. Model (4) is a binary probit for the probability of being above the poverty line, where household living below poverty line = 0 and above poverty line = 1. Table 4 reports the coefficient estimates for the probit models and marginal effects are reported in the Appendix, Table A1.

The following control variables are included in all of these models: education level of the household head (measured as completed years of education); dummy variables for the occupation status of the household head: high (business and service), medium (farming, small business, and boatman), and low (day labor, rickshaw/van pulling, fishing, household-based self-employment and begging), which is the reference category; number of individuals from the household living abroad (who tend to send remittances to the family in Bangladesh); amount of owned cultivable land (in decimals[19]); months of duration of membership in the microcredit program; and age- and sex-adjusted household size. We did not use the latter in model (i) and model (ii) because the dependent variables of both models were measured in age- and sex-adjusted context. In addition, we have controlled for main earning source of the villagers, which is a dummy variable (predominantly business = 1, predominantly farming = 0). Our expectations are that all of the control variables, other than age- and sex-adjusted household size, will have a positive association with each of the dependent variables.

[18]In further analyses we also applied robust regression and quantile regression to these models and the results do not differ significantly from the OLS results reported here.
[19]Decimals are a standard unit of area in Bangladesh; 100 decimals are equal to one acre.

TABLE 4
Regression Results

	(i) OLS	(ii) OLS	(iii) Ordered Probit[a]	(iv) Probit[a]
	Log of (Annual) per Capita Income	Log of per Capita Nonland Assets	Food Sufficiency (Deficit = 0, Neutral = 1, Surplus = 2)	Poverty Status (0 = Below Poverty Line, 1 = Above)
Education of the household head (years)	0.031*** (0.008)	0.029* (0.017)	0.099*** (0.023)	0.083*** (0.032)
Household head in medium-grade occupation[b]	0.151*** (0.059)	0.244* (0.132)	0.077 (0.173)	0.330* (0.198)
Household head in high-grade occupation[b]	0.281*** (0.080)	0.493** (0.205)	0.603*** (0.242)	1.467*** (0.520)
No. of persons from household living abroad	0.452*** (0.080)	0.588*** (0.123)	0.376* (0.212)	1.002*** (0.406)
Owned cultivable land (decimals)	0.002*** (0.001)	0.004*** (0.001)	0.007*** (0.002)	0.010*** (0.004)
Membership in GB microcredit (months)	0.001** (0.0003)	0.002** (0.001)	0.004*** (0.001)	0.001 (0.001)
Age- and sex-adjusted household size	–	–	−0.084 (0.068)	−0.258*** (0.087)
Main earning source of the villagers (1 = business, 0 = farming)	0.251*** (0.071)	0.130 (0.152)	0.104 (0.210)	0.095 (0.279)
MHI status (1 = established MHI, 0 = no MHI)	0.080 (0.069)	0.096 (0.162)	0.355* (0.205)	0.202 (0.246)
Constant	5.090*** (0.052)	4.184*** (0.147)	–	0.790** (0.350)
Cut point 1	–	–	0.126 (0.286)	–
Cut point 2	–	–	1.838 (0.309)	–
Observations	244	244	244	244
F-statistics	27.05***	10.24***	–	
Wald chi-squared	–	–	69.51***	33.42
R^2	0.40	0.20	–	–
Pseudo R^2	–	–	0.14	0.19
Reset	$F(3, 232) = 0.81$ Prob > $R^2 = 0.489$	$R^2(3, 232) = 0.76$ Prob > $R^2 = 0.515$	$\chi^2(1)$ $R^2 = 0.55$ Prob > $\chi^2 = 0.457$	$\chi^2(1)$ $R^2 = 0.69$ Prob > $\chi^2 = 0.407$

Notes: *** indicates significant at the 1% level, **indicates significant at the 5% level, and *indicates significant at the 10% level. Figures in parentheses are Huber–White robust standard errors.
[a]Low grade occupation is the reference category.
[b]Coefficients reported for probit and ordered probit models; marginal effects are shown in Appendix, Table A1.

Looking across the columns in Table 4, the coefficient estimates of entire variables have the expected signs in all of the models, and each variable affects at least one outcome. There are four variables that appear to consistently contribute positively to the four outcome measures; these are education level of the household head, household head having a high-grade occupation, the number of household members living abroad, and the amount of cultivable land owned by the household. The household head having a medium-grade occupation and the length of membership in the microcredit scheme affect three of the four outcomes, and household size and main earning source of the villagers affect one outcome. Quantitatively the household head having a high-grade occupation and amount of owned cultivable land make the largest contribution to all four outcomes; for example, from Table A1 in the Appendix we see that the household head having a high-grade occupation increases the probability of being in food surplus by 0.167 percentage points and increases the probability of being above the poverty line by 0.206 percentage points. In all cases the coefficient on MHI placement status is positive and it is statistically significant in determining food sufficiency. The marginal effects in Table A1 show that being in an area where MHI has been available for at least 5 years increases the chances of being in food surplus by 0.078 percentage points. This is a relatively large effect, being similar in size to having another person from the household living abroad and second only to the household head having a high-grade occupation.

In terms of the diagnostic statistics, the models appear to be well specified; each of them is jointly significant at the 1 percent level according to the *F*-tests and Wald tests and there is no evidence of misspecification according to the Reset tests. The explanatory power of the models is respectable for this type of cross-section analysis with primary data. In addition to exploring the possible collinearities between the explanatory variables, each variable was excluded in turn and this made little difference to the existing coefficient estimates. We also measured pairwise correlations (see Table A2 in the Appendix) and variance inflation factors (VIFs). There are a small number of correlations significant at 10 percent, but in all cases the variables involved are individually statistically significant in the models. In all cases the VIFs (not reported) are below the tolerance limit, suggesting that multicollinearity is not a severe problem in our models.

DISCUSSION

Our results suggest that MHI placement has a significant beneficial effect on food sufficiency but is not significant in contributing to our other poverty indicators. There are a number of possible reasons for finding a statistically insignificant association with income, nonland assets, and the probability of being above the poverty line. First, MHI may take time to generate a significant impact on household income via improvements in health status, and thus the dynamic effects may not have been observed in our cross-section analysis, where the average time of program placement in GB1 was only 6 years. It is reasonable to expect that food sufficiency may react before the ownership of nonland assets and rising above the poverty line, which rely on an accumulation of surplus income over time; however, via these dynamic arguments we may expect per capita incomes to rise relative quickly as a result of MHI placement, and our evidence suggests they do not. Second, the result may be due to the type of insurance and health care provided through this MHI scheme.

GB does not provide secondary or tertiary health care from its health centers, and although it maintains a referral mechanism for the higher levels of health care, this does not function effectively as we observed during the survey.[20] As was seen in Table 2, health care expenditure and workdays lost (the main determinants of stabilization of household income in Figure 1) did not differ significantly between the program area (GB1) and comparison area (GB3). The adverse effects of protection against moral hazard may be a possible reason for the former. Copayments are used to deter overutilization of health care by the insured, but qualitative information collected during our survey revealed that these copayments, although relatively small, are a strong deterrent to these very poor households members, who then prefer to use informal health care providers who may offer credit terms, even if the total cost of health care is then higher.

Of course one possible further explanation is that in fact the addition of MHI to microcredit does not contribute to most poverty indicators. However, we should be wary before reaching this conclusion because while our study uses primary data collected expressly for assessing the impact of MHI and applies appropriate methods for analyzing the data, there are still some limitations that largely arise from time and resource constraints.[21] First, although panel data or repeated cross-section data are the most suitable for analyzing dynamic outcomes, our study uses a single cross-section of households. In addition, our study does not provide evidence on whether MHI can reach the poorest of the poor because we used a sample of microcredit members and not the population as a whole. It has been shown that microcredit is not always able to reach the ultra poor (Amin, Rai, and Topa, 2003; Navajas et al., 2000), but they are clearly a key group for policy concerns.

CONCLUSIONS

This study has outlined the causal mechanisms by which adding MHI to microcredit schemes can contribute to improving the poverty status of households. We have evaluated this in the context of Grameen Bank, the largest microcredit organization in Bangladesh. Our analyses clearly demonstrate the positive effect of microcredit in reducing poverty, while the effect of adding micro health insurance is still too new to be able to offer definitive results. The associations between the availability of micro health insurance and three poverty measures are negative (i.e., a reduction in poverty), yet only statistically significant for food sufficiency.

A number of reasons have been suggested for our findings, which include problems in detecting long-term effects with our cross-section data as well as shortcomings of the MHI scheme in question, including a lack of proper referral services and the adverse effects of protection against moral hazard. Given the former problem, it may not be sensible to draw generalized conclusions from our findings. Nevertheless, we

[20] Officially there is a provision that an insured household will receive up to TK.2000 (about US$29) annually as referral (hospitalization) benefits. This benefit was provided via external funding from the ILO. However, the fund has been exhausted so the benefit is no longer provided in practice.

[21] There was no external research funding for this study.

are encouraged by these results and suggest to public policymakers that continued collection of usable data for future evaluation be considered a critical part going forward. A similar analysis, with more years of experience and therefore better data, needs to be conducted in the years ahead.

APPENDIX

TABLE A1
Marginal Effects for the Ordered Probit and Probit Models in Table 4

	Food Sufficiency Level			Poverty Status (0 = Under Poverty Line, 1 = Above)
	Food Deficit = 0	Neither Deficit nor Surplus = 1	Surplus = 2	
Education of the household head (years)	−0.032***	0.010**	0.022***	0.021***
	(0.008)	(0.004)	(0.006)	(0.008)
Household head in medium-grade occupation[a]	−0.025	0.008	0.017	0.084*
	(0.056)	(0.018)	(0.038)	(0.050)
Household head in high-grade occupation[a]	−0.165***	−0.002	0.167**	0.206***
	(0.055)	(0.031)	(0.077)	(0.037)
No. of persons from household living abroad	−0.123*	0.039	0.084*	0.253***
	(0.069)	(0.026)	(0.048)	(0.098)
Owned cultivable land (decimals)	−0.002***	0.001**	0.001***	0.003***
	(0.001)	(0.0003)	(0.0004)	(0.001)
Membership in GB microcredit (months)	−0.001***	.0004**	0.001***	0.0004
	(0.0003)	(0.0002)	(0.0002)	(0.0003)
Age- and sex-adjusted household size	0.028	−0.009	−0.019	−0.065***
	(0.022)	(0.008)	(0.015)	(0.022)
Main earning source of the villagers (1 = business, 0 = farming)	−0.033	0.010	0.024	0.024
	(0.067)	(0.018)	(0.049)	(0.068)
MHI status (1 = established MHI, 0 = no MHI)	−0.117*	0.039	0.078*	0.051
	(0.068)	(0.026)	(0.045)	(0.064)

Note: Marginal effects for dummy variables evaluated for discrete a change from 0 to 1. ***indicates significant at the 1% level, ** significant at the 5% level, and * significant at the 10% level. Figures in parentheses are Huber–White robust standard errors.
[a]Low-grade occupation is the reference category.

TABLE A2
Pairwise Correlations Between the Control Variables

	Education of Household Head (Years)	Household Head in Medium-Grade Occupation	Household Head in High-Grade Occupation	No. of Persons From Household Living Abroad	Owned Cultivable Land (Decimals)	Membership in GB Microcredit (Months)	Age- and Sex-Adjusted Household Size	Main Earning Source of the Villagers (1 = Business With Farming, 0 = Farming)
Education of the household head (years)	1							
Household head in medium-grade occupation	0.033	1						
Household head in high-grade occupation	0.039	−0.390*	1					
No. of persons from household living abroad	−0.069	0.049	0.011	1				
Owned cultivable land (decimals)	0.001	0.078	0.072	0.124	1			
Membership in GB microcredit (months)	0.044	0.156*	0.058	0.078	0.066	1		
Age- and sex-adjusted household size	−0.023	0.067	0.109	−0.081	0.169*	0.204*	1	
Main earning source of the villagers (1 = business with farming, 0 = farming)	−0.009	0.035	0.102	0.103	0.024	0.218*	−0.023	1

* indicates significant at the 10% level.

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